Benchmarking Approximate Nearest Neighbor Search in Real-time Applications

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Abstract

Real-time query processing over high-dimensional vector databases poses significant computational challenges for latency-sensitive applications. This study presents a rigorous empirical benchmark comparing HNSW, IVF-PQ, and graph-based indexing strategies across varied hardware configurations. Our findings reveal critical memory bandwidth bottlenecks and suggest optimal indexing parameterizations for microsecond-scale retrieval in large-scale search infrastructure.

Keywords: Information Retrieval, Database Systems, Benchmarking, Distributed Systems

1. Introduction

With the proliferation of deep embedding representations across web search, recommender systems, and scientific indexing, Approximate Nearest Neighbor (ANN) search algorithms serve as a critical infrastructure layer. Despite numerous publications claiming superior Pareto frontiers, systematic cross-architecture evaluations remain sparse.

2. Evaluation Methodology

We standardise the evaluation framework using identical CPU instruction sets, memory pinning, and thread allocation across cosine distance, Euclidean metric, and inner-product spaces. Datasets range from 128 to 1536 dimensions.

3. Memory Hierarchy Analysis

Cache locality analysis highlights that graph-based approaches incur significant random memory accesses during beam traversal. We propose hardware-aware prefetching routines to alleviate cache thrashing.

References

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